

1. Intro

Period: 2

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Group Name: Gals

Project Title: Tom and Jerry(s)

2. Description

Describe your project:

- A single-user game where the user (cat) tries to catch as many mice as possible. There are multiple levels that the user can advance to. This game is similar to whack-a-mole, with the mice periodically disappearing and reappearing. The user is given a set time to reach a goal number of mice caught (number will increase with each level, time will decrease with each level). If the user does not reach the goal, the user loses the game. If the user catches the goal number of mice within the time limit, they are allowed to continue catching mice (this will contribute to their high score).

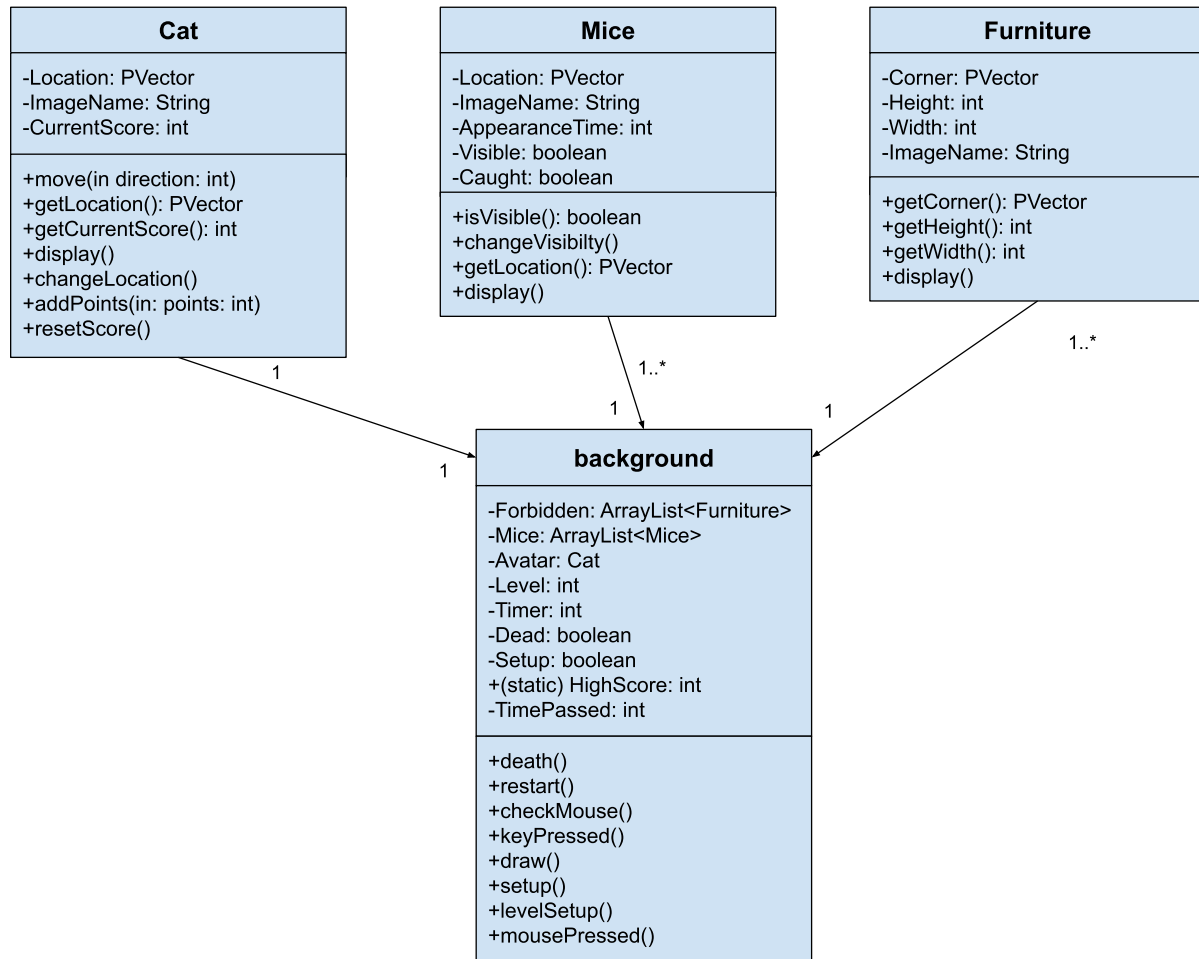
List and explain functionalities:

- Arrow keys: user uses arrow keys to navigate the cat, allows user to move up/down/left/right
- Background: consists of a grid (house) where mice will pop up, mice's positions will be randomly generated; the user will lose the game if they touch certain furniture
- Timer: different time limits will be set per level
- Scoreboard: keeps track of number of mice caught

Any libraries needed, indicate the name and how you are using it in your project:

- Timer? → need to research how to add in a timer

3. UML Diagram



4. How does it work?

The user will assume the role of a cat (which will appear on screen) and try to catch as many mice as possible. At each level, the user will use the arrow keys to navigate their cat to catch mice across the screen, with the mice being caught when the cat reaches their position. The user will need to avoid furniture throughout the screen as they make their way towards catching mice. Mice will periodically disappear and reappear until caught, similar to the game of whack-a-mole. The user is given a set time to reach a goal number of mice caught: the goal will increase and the time given will decrease as the player advances levels. If the user does not reach the goal or touches any furniture, the user loses the game. If the user catches the goal number of mice within

the time limit, they are allowed to continue catching mice (this will contribute to their high score). A scoreboard will display the user's current score and high score and a timer will display the user's remaining time for the level.

5. Functionalities/Issues **(bolded text are new changes)**

- Background.pde:
 - Current: instance variables, setup (sets up initial values for variables), **death (checks if the timer ran out or if the cat touched furniture), restart (initiates the game state that allows the restart screen to be displayed, resets variables), checkMouse (checks if the cat touched a mouse), keyPressed (allows the user to move), levelSetup (so far, has the setup for the start screen, restart screen, and the map for level 1), mousePressed (allows the user to pressed start and restart buttons), draw**
 - By next meeting: **finish setting up the maps remaining 2-3 levels (set the backgrounds and locations for furniture and mice)**
- Cat.pde:
 - Current: instance variables, constructor, move (increments the cat's x/y value based on the inputted direction), getLocation (returns cat's location as a PVector), getCurrentScore (returns the current score of the cat), display (displays the cat avatar at its location)
 - By next meeting: N/A (complete)
- Furniture.pde:

- Current: instance variables, constructor, getCorner (returns furniture's location as a PVector), getHeight (returns furniture's height), getWidth (returns furniture's Width), display (displays the furniture avatar at its location)
- By next meeting: N/A (complete)
- Mice.pde:
 - Current: instance variables, constructor, isVisible (returns if the mouse is visible), **changeVisibility (makes mouse invisible if already visible and visible if invisible)**, getLocation (returns the location of the mouse), display (displays the mouse avatar if visible)
 - By next meeting: N/A (**complete**)
- Issues:
 - Merging branches: both Amanda and Stuti were having trouble with pushing their changes onto the main branch (due to merge conflicts, as well as general issues with pushing) → we solved this issue by consulting the apcs repo, where we followed the instructions given in the classwork document to solve our issues
 - **We had an issue with checking if the cat was at the same location as the furniture (the cat would die even if it was far away from the furniture, or it wouldn't die if it was already touching it) → we resolved this issue by realizing that we had to add an offset for how big the cat is (since the location is the top left corner)**